Project 2: Basic Encryption & Decryption

Caesar Cipher — DecodeLabs Cyber Security Internship (Batch 2026)
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1. Objective

The goal of this project is to implement a simple, reversible encryption and decryption technique using the Caesar Cipher. This demonstrates the core concept of data confidentiality: transforming readable plaintext into unreadable ciphertext, and reliably reversing that transformation using the same key.

2. Approach

The Caesar Cipher works by shifting each letter of the alphabet forward (encryption) or backward (decryption) by a fixed number of positions, known as the shift key (n). Characters are converted to their ASCII values with ord(), shifted mathematically, wrapped using modulo 26 so the shift stays within the alphabet, and converted back to letters with chr(). Non-alphabetic characters such as spaces and punctuation are passed through unchanged.

Encryption formula: $E(x) = (x + n) \bmod 26$

Decryption formula: $D(x) = (x - n) \bmod 26$

3. Implementation

The program is written in Python and consists of two core functions, encrypt() and decrypt(), plus a simple text menu that lets the user choose to encrypt, decrypt, or run a full round-trip test. Uppercase and lowercase letters are handled separately so that letter case is preserved in the output.

4. Test Results

Message	Shift Key	Encrypted	Decrypted
HELLO	3	KHOOR	HELLO
Cyber Security	5	Hdgjw Xjhzwnyd	Cyber Security

In both test cases, decrypting the ciphertext with the same shift key returned the exact original message, confirming that the encryption and decryption logic are correct and symmetric.

5. Limitations & Observations

- Small key space: only 25 possible shift keys, making brute-force attacks instant.
- Pattern preservation: letter frequency patterns in the ciphertext match plaintext, making the cipher vulnerable to frequency analysis.

- The Caesar Cipher is suitable for learning core cryptographic concepts, but is not secure for real-world use. Modern systems use algorithms such as AES, which rely on much larger key spaces and additional techniques like confusion and diffusion.

6. Conclusion

This project successfully implements a working Caesar Cipher encryption and decryption tool in Python, validated with multiple test cases. It builds a foundational understanding of cryptographic logic, ASCII manipulation, and modular arithmetic that underpins more advanced encryption systems used in professional cybersecurity practice.